



E-commerce Sales Dashboard

Data Storytelling & Statistical Validation

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Project Objective



Data Analysis

Comprehensive analysis of e-commerce sales patterns and customer behavior



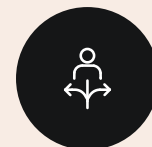
Pattern Recognition

Identification of key trends, seasonal patterns, and revenue drivers



Interactive Dashboard

Development of an intuitive Power BI dashboard for real-time insights



Strategic Support

Empowering data-driven decision-making for business growth

Executive Summary

- Analyzed 166K orders and 3K customers
- Built interactive Power BI dashboard
- Identified key trends and revenue drivers
- Enabled data-driven business insights



Key Performance Indicators

3.87M

Total Revenue

Aggregate sales across all periods and regions

166K

Total Orders

Complete transaction count in the dataset

23.23

Average Order Value

Mean revenue per transaction

3K

Total Customers

Unique customer base analyzed

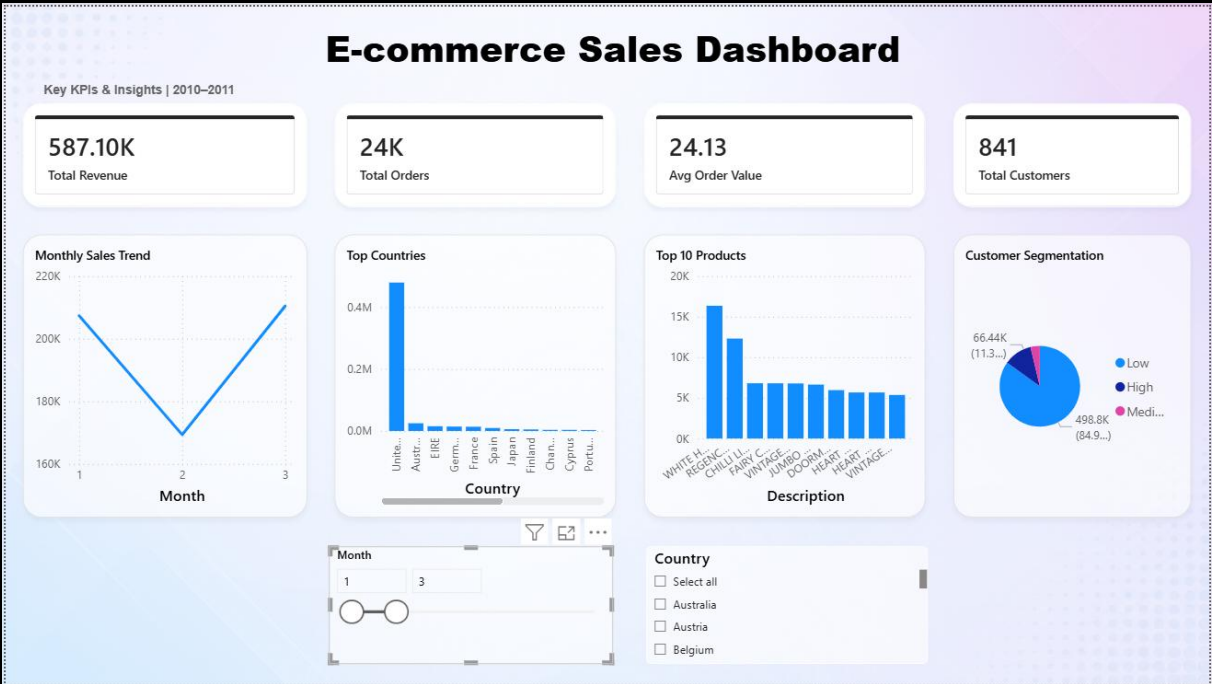
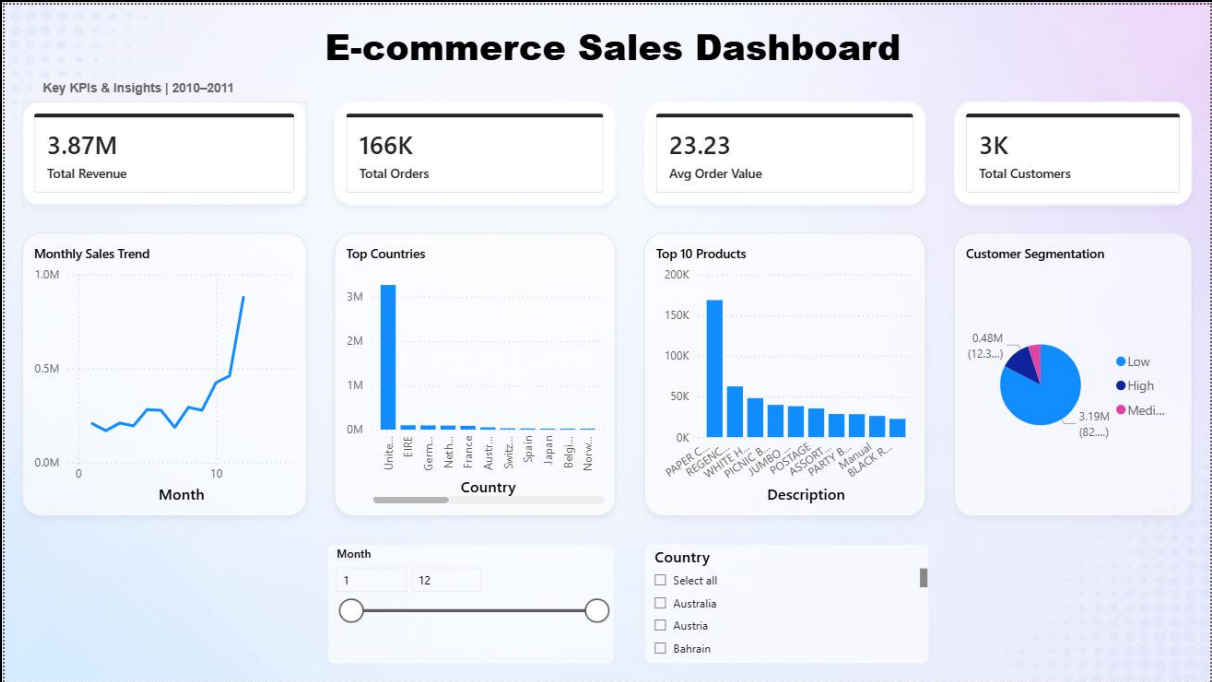
Dashboard Features & Capabilities

Visual Components

- Monthly sales trend analysis with year-over-year comparison
- Top countries by revenue contribution and geographic distribution
- Product performance breakdown with top 10 products highlighted
- Customer segmentation by value tiers and engagement levels

Interactive Elements

- Dynamic month selection filter for time-based analysis
- Country filter for regional performance comparison
- Real-time calculations and aggregated metrics
- Drill-down capabilities for detailed investigation



Key Insights Uncovered

1

Revenue Growth

Revenue demonstrates strong growth trajectory in later months, with Q4 showing significant increase compared to earlier quarters. This indicates seasonal shopping behavior.

2

UK Dominance

United Kingdom generates the highest revenue among all countries, establishing it as the primary market for business focus and investment.

3

Top Products

Top 10 products contribute disproportionately to total sales, following the 80/20 rule where few products drive majority of revenue.

4

Customer Segmentation

Majority of customers fall under low-value segment, presenting opportunity for engagement improvement and value enhancement strategies.

Business Interpretation



Seasonal Nature

Sales exhibit clear seasonal patterns with peak performance in Q4 holiday season



Geographic Focus

UK market represents primary revenue source and strategic expansion opportunity



Product Strategy

Concentration of revenue in top products suggests focused promotion approach



Customer Engagement

Low-value customer segment indicates need for retention and upselling initiatives



Strategic Recommendations



Q4 Marketing Campaigns

Amplify marketing investments during Q4 peak season to capitalize on established seasonal trends



UK Market Expansion

Prioritize geographic expansion and market penetration strategies in United Kingdom



Product Promotion

Focus promotional efforts on top-performing products to maximize revenue impact



Customer Loyalty Programs

Introduce targeted engagement programs for low-value customers to increase lifetime value

Hypothesis Testing Methodology

Research Hypothesis

Null Hypothesis (H_0): No difference in monthly revenue

Alternative Hypothesis (H_1): Revenue increases in later months

01

Hypothesis Formulation

Defined testable claim about seasonal revenue patterns

03

Pattern Validation

Confirmed higher Q4 revenue supports seasonal trend existence

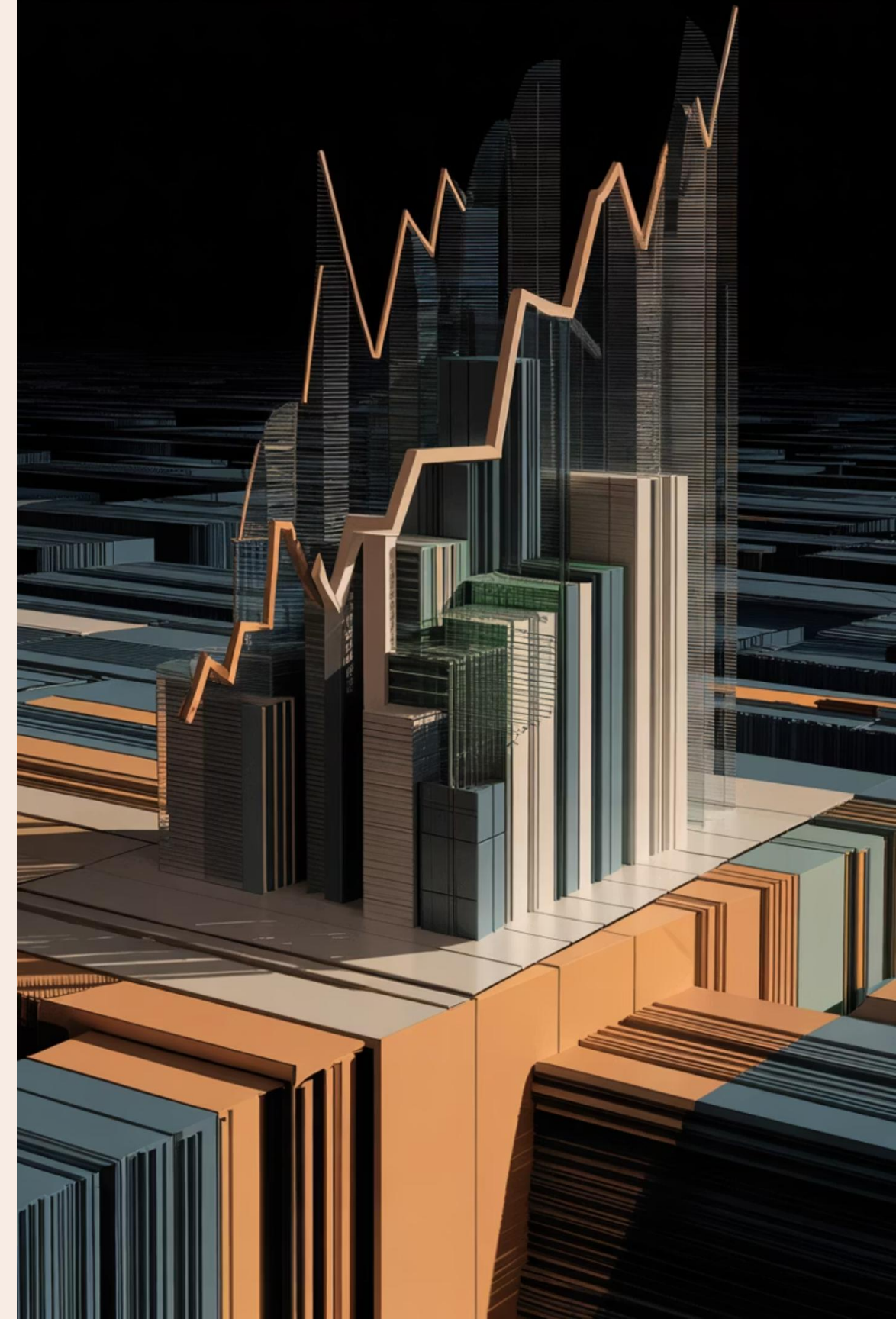
Statistical Analysis

Revenue data analysis revealed higher Q4 values compared to earlier quarters, supporting the seasonal trend hypothesis.

02

Data Observation

Analyzed revenue distribution across all time periods



Conclusion & Impact



Data-Driven Decision Making

Transformed raw data into actionable business insights that inform strategic planning



Effective Visualization

Leveraged Power BI's capabilities to create intuitive, interactive dashboards for storytelling



Growth Opportunities

Identified specific areas for revenue optimization and customer engagement improvement

This project demonstrates my ability to transform raw data into meaningful business insights using data analysis and visualization. It highlights how data-driven approaches can support informed decision-making and uncover valuable growth opportunities.